Name: _____

WORKSHEET

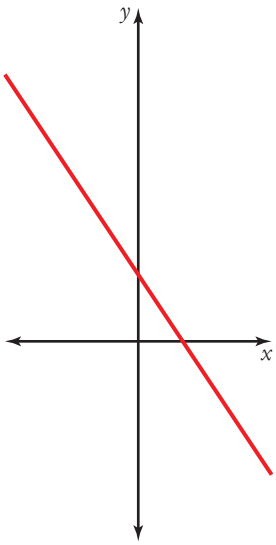
Gradient functions

1 For the following functions:

i State where they are increasing decreasing or stationary.

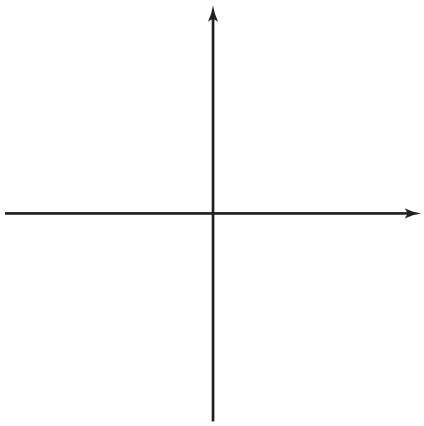
ii Sketch the gradient function.

a

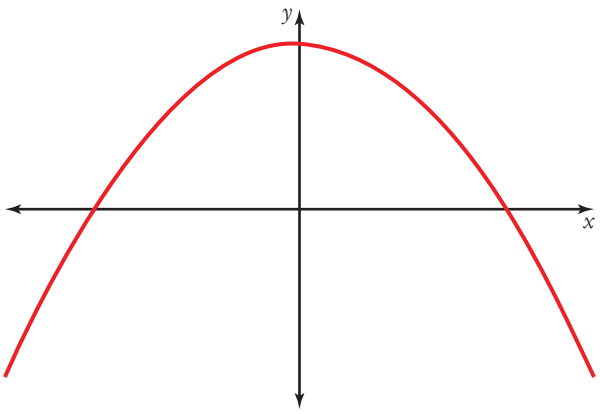


i _____

ii

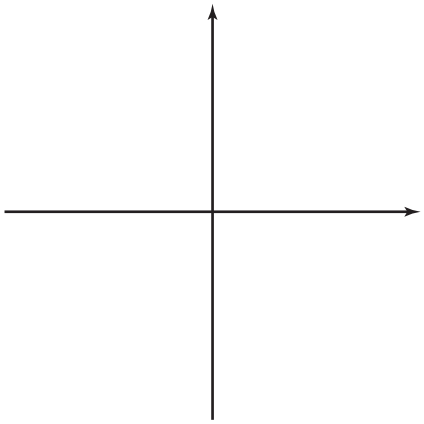


b



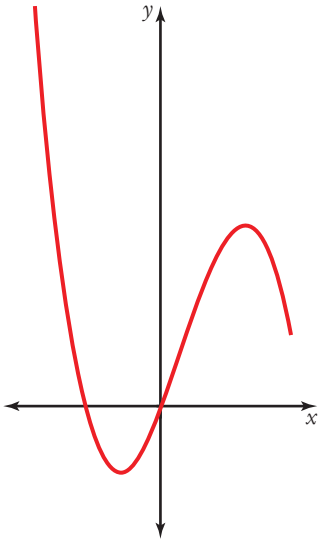
i _____

ii



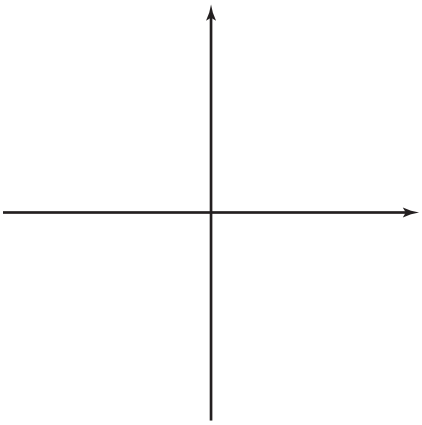
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c

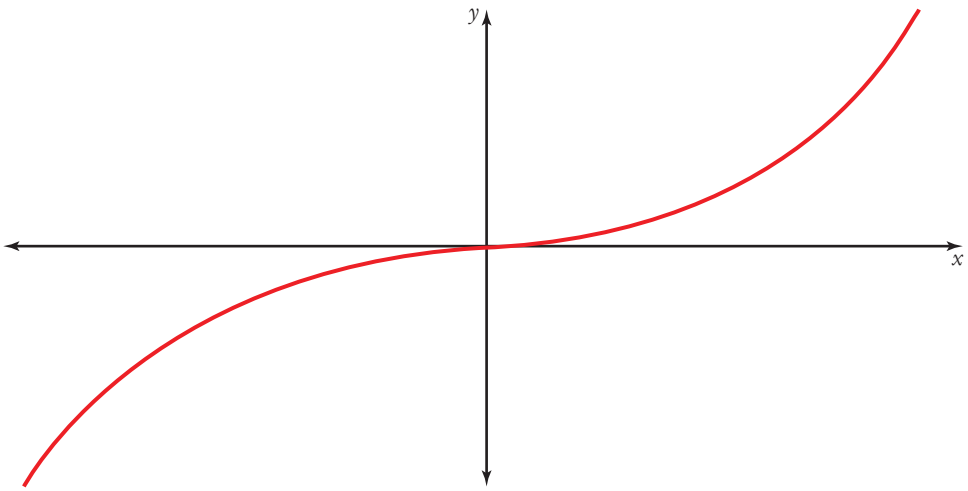


i _____

ii

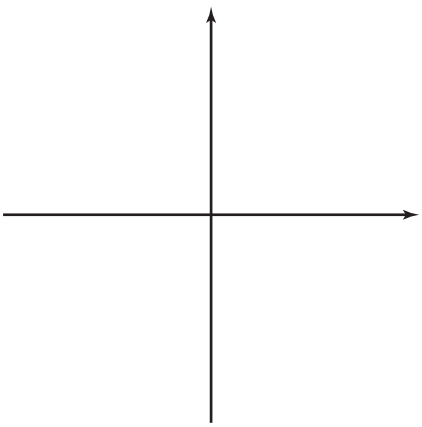


d

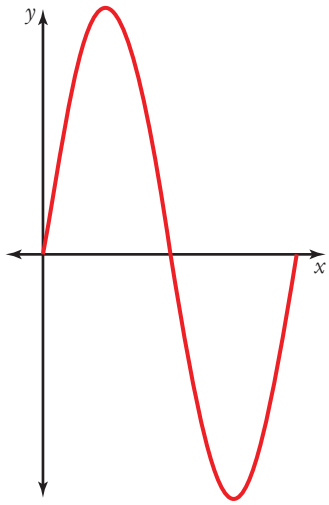


i _____

ii

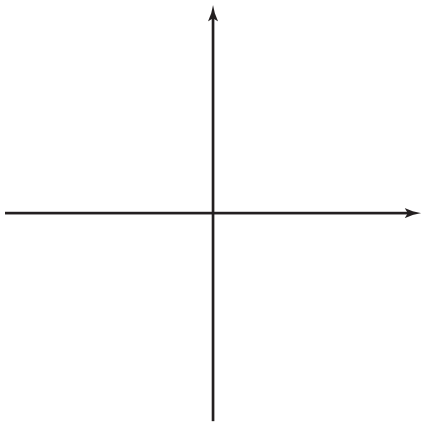


e Only consider $0 \leq x \leq 2\pi$

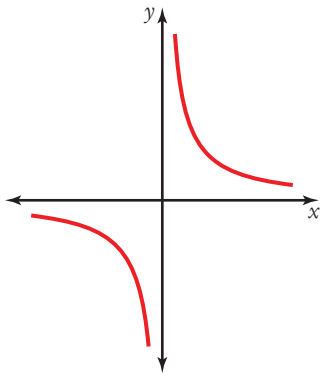


i _____

ii

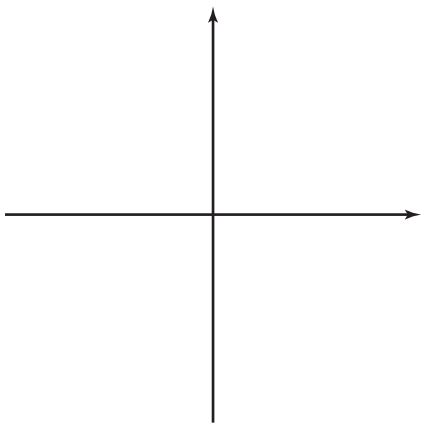


f



i _____

ii

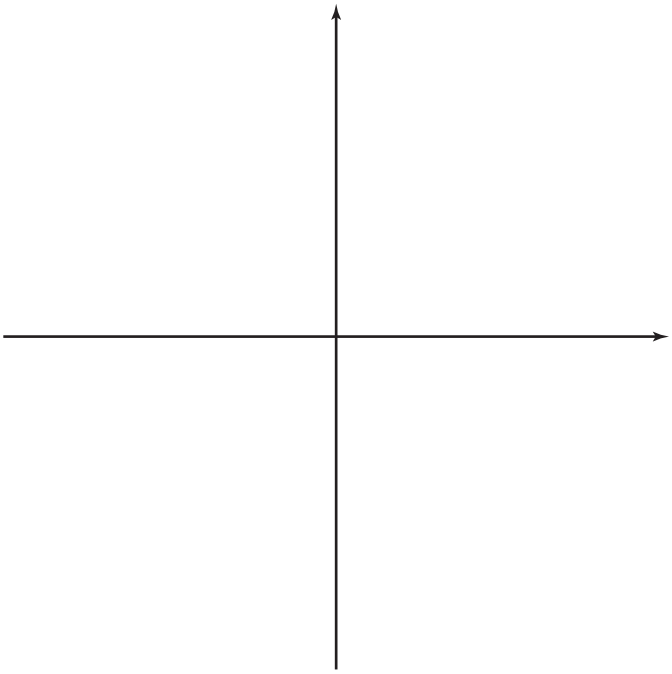


2 The derivative of a function can be used to determine the nature of the function (where it is stationary, increasing or decreasing).

a Given, $f'(x) = 4x^3 - 3x^2 - 20x + 19$ complete the table.

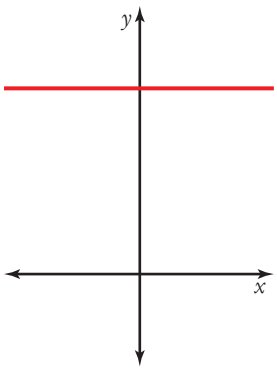
x	-3	-2	-1	0	1	2	3
$f'(x)$							
Nature of $f'(x)$							

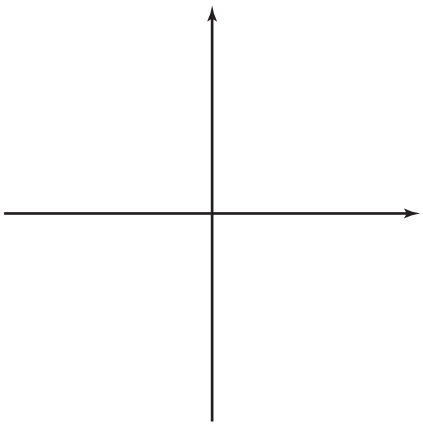
b Sketch the gradient function $f'(x) = 4x^3 - 3x^2 - 20x + 19$.



3 Sketch an example of a function that would have the following gradient function.

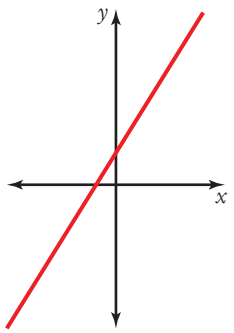
a

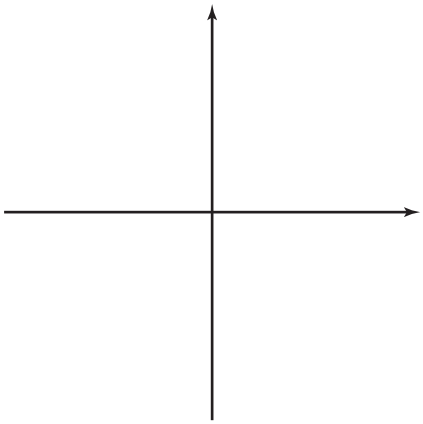




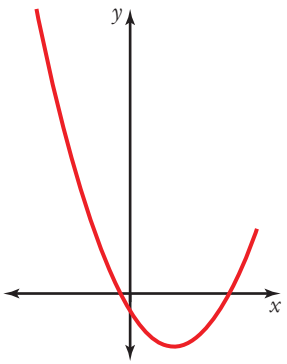
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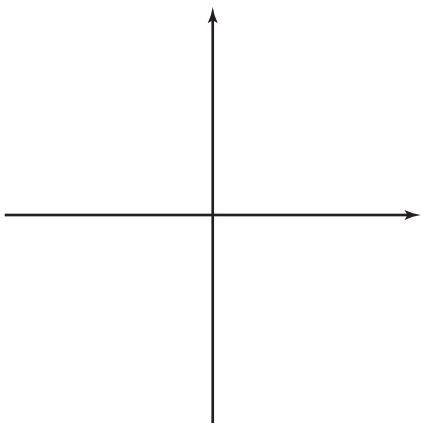
b





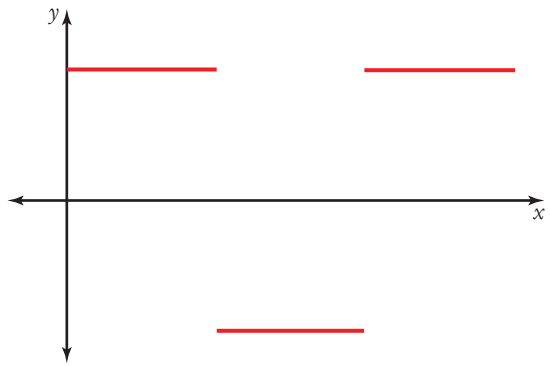
c

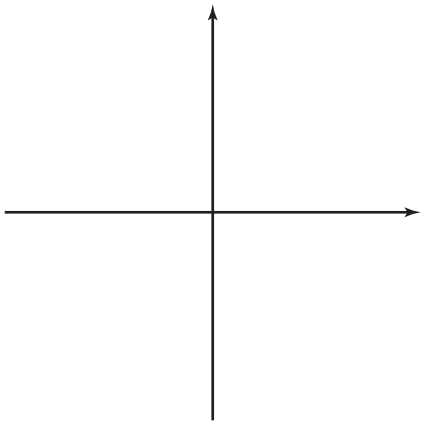




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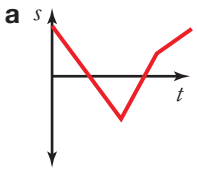
d



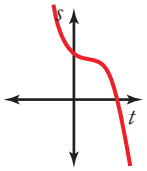


4 The graphs below show some displacement–time and velocity–time graphs for some moving objects. For each displacement-time graph, choose the most likely velocity–time graph. Write A, B, C or D underneath each of the first graphs to show which is the matching graph from below.

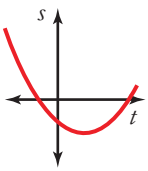
a



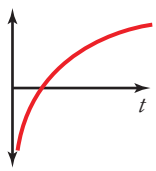
b



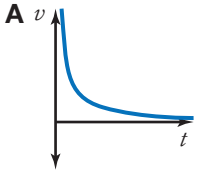
c



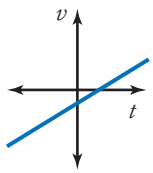
d



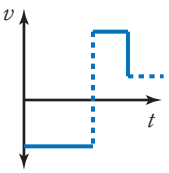
A



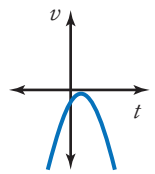
B



C



D

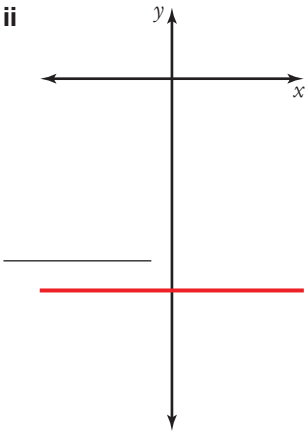


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Answers

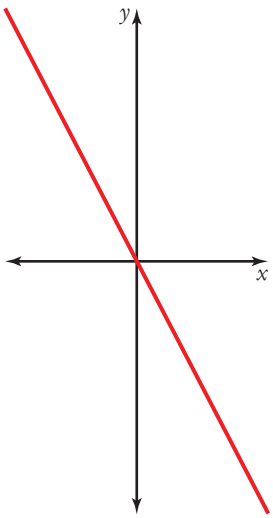
1 a i decreasing for all $x \in (-\infty, \infty)$

ii



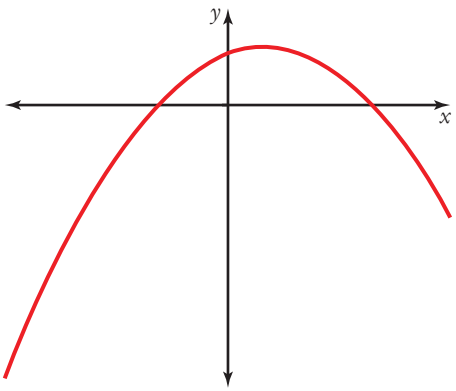
b i increasing for $x < 0$, stationary at $x = 0$ and decreasing for $x > 0$

ii



c i increasing for $a < x < b$, stationary at $x = a, b$ and decreasing for $x < a, x > b$

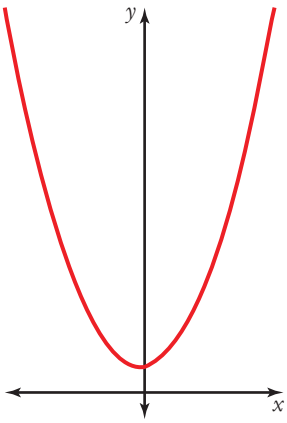
ii



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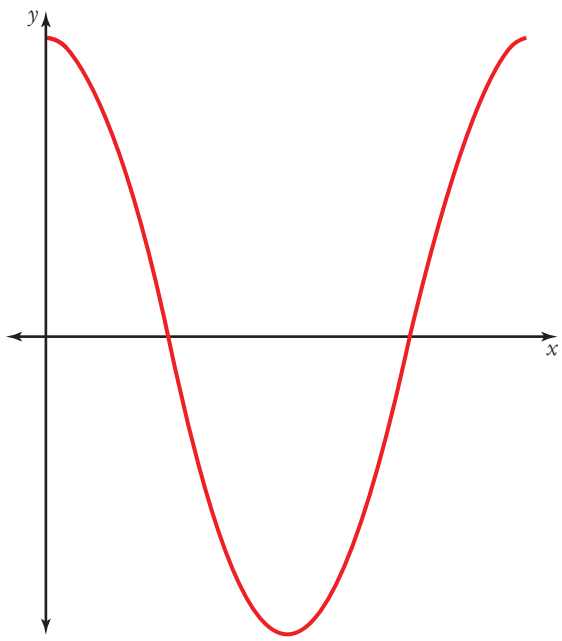
d i increasing for all $x \in (-\infty, \infty)$

ii



e i increasing for $0 \leq x < \frac{\pi}{2}$, $\frac{3\pi}{2} < x \leq 2\pi$, stationary at $x = \frac{\pi}{2}$, $\frac{3\pi}{2}$, decreasing for $\frac{\pi}{2} < x < \frac{3\pi}{2}$

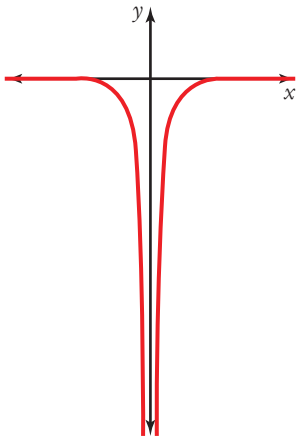
ii



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f **i** decreasing for all $x < 0$, $x > 0$. Note that $x \neq 0$.

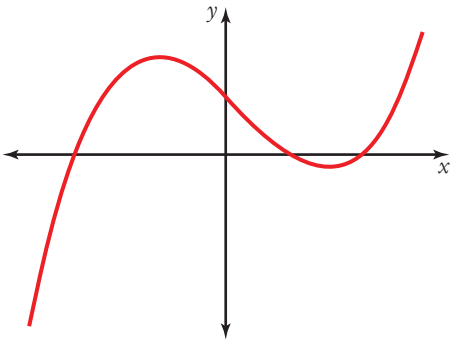
ii



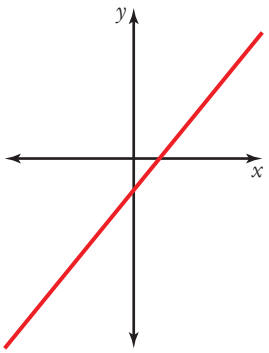
2 **a**

x	-3	-2	-1	0	1	2	3
$f'(x)$	-56	15	32	19	0	-1	40
Nature	Decrease	Increase	Increase	Increase	Stat.	Decrease	Increase

b

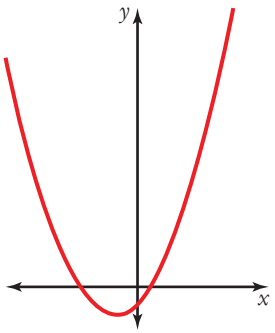


3 **a**

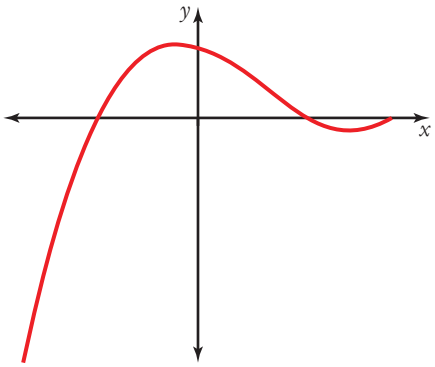


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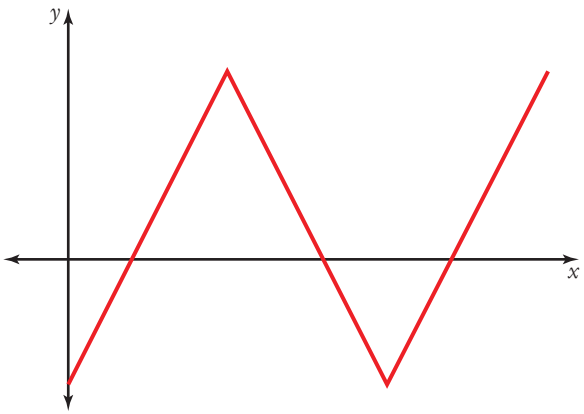
b



c



d



4 **a** graph C

b graph D

c graph B

d graph A

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